

S-Entropy Embedding Uniqueness and the Forced Measurement Protocol: Deriving the State Encoder from Bounded Phase Space Theory

Kundai Farai Sachikonye

Chair of Experimental Bioinformatics, Technical University of Munich

AIME Registry for Artificial Intelligence

Maximus-von-Imhof-Forum 3, 85354 Freising, Germany

kundai.sachikonye@wzw.tum.de

May 2026

Abstract. We derive the three-coordinate S-entropy state encoder as a mathematical necessity of any observer operating on bounded phase space systems, not as an engineering design choice. The argument proceeds from the Bounded Phase Space axiom through five theorems to a unique forced measurement protocol. Poincaré recurrence forces all persistent bounded systems into oscillatory regimes with discrete Koopman spectra. Three scalar functionals of this spectrum — kinematic entropy S_k , temporal span S_t , and harmonic fraction S_e — are shown to be the minimal sufficient statistics for categorical system identification under the operations supported by the membrane computing substrate. The map $\sigma: \mathcal{X} \rightarrow [0, 1]^3$, $\sigma(X) = (S_k, S_t, S_e)$, satisfies a categorical sufficiency theorem: two systems with identical S-entropy coordinates are computationally indistinguishable under aperture filtering, cascade routing, and strobe readout. A timescale separation theorem establishes that S_k , S_t , and S_e must be measured on hierarchically ordered timescales $\tau_k \ll \tau_t \ll \tau_e$, each derived from the system’s own spectral structure. The measurement sequence is therefore forced: any observer that violates the timescale ordering generates aliased coordinates that fail the sufficiency condition. The S-entropy space carries a natural product Fisher information metric $g = g_k \oplus g_t \oplus g_e$ compatible with the partition depth measure from the bilayer substrate. A conservation law $\sum_k p_k = 1$ holds on the spectral probability simplex by Parseval’s theorem; this law is preserved under the three-pass measurement protocol. Validation on 39 standard compounds confirms numerical separability: the six chemical families occupy disjoint regions of S-entropy space, with inter-family distances 1.09 to 4.38 times larger than intra-family diameters.

Keywords: S-entropy, Koopman spectrum, state encoder, forced measurement, categorical sufficiency, Fisher information, partition coordinates, Parseval conservation, timescale separation, bounded phase space

1. Introduction

Every computing substrate requires a state encoder: a procedure that maps the physical state of an input system to a representation on which the computer’s operations act. For classical digital systems, the encoder is the analogue-to-digital converter, whose design is guided by the Nyquist-Shannon sampling theorem [Shannon and Weaver, 1949]. For neuromorphic and analogue systems, the encoder is typically designed by heuristic opti-

misation against a training corpus [Maass, 1997]. Neither approach is derived from first principles; both presuppose a fixed target representation and engineer backward to the encoder.

This paper derives the encoder forward from the physics of bounded phase space. The starting point is purely mathematical: any persistent physical system occupies a bounded region of phase space with finite Liouville measure [Liouville, 1838]. From this single axiom, we show that the system’s Koopman spectrum [Koopman, 1931] is the unique

information-complete representation of its state, and that three scalar functionals of this spectrum — kinematic entropy S_k , temporal span S_t , and harmonic fraction S_e — are the minimal sufficient statistics for categorical identification compatible with the membrane computing operations established in the companion bilayer-substrate paper [Sachikonye, 2026a].

The three-coordinate system we derive is not a design choice but a forced consequence of three constraints operating simultaneously: (i) completeness under categorical operations, (ii) measurability without demolishing the partition structure of the substrate, and (iii) computability on the membrane hardware without external reference signals. We call this the *forced measurement protocol*: once the Bounded Phase Space axiom is accepted and the membrane substrate is fixed, every other encoding is either informationally incomplete or destroys the substrate’s partition structure.

The paper is structured as follows. Section 2 establishes the Koopman spectral representation from the Bounded Phase Space axiom. Section 3 defines the three S-entropy coordinates and establishes their basic properties. Section 4 proves the categorical sufficiency theorem. Section 5 establishes the timescale separation and the non-demolition commutation relation. Section 6 derives the forced measurement protocol. Section 7 proves the S-entropy conservation law and the Fisher metric structure. Section 8 describes the physical realisation of the three passes in the membrane substrate. Section 9 presents numerical validation. Sections 10 and 11 discuss and conclude.

2. Bounded Phase Space and Koopman Spectra

2.1. The Foundational Axiom

Axiom 1 (Bounded Phase Space). Every persistent physical system $\mathcal{X}$ occupies a region $\Omega \subset \mathbb{R}^{2d}$ of d -dimensional phase space satisfying:

$$\mu(\Omega) = \int_{\Omega} \frac{d^d q d^d p}{h^d} < \infty, \quad (1)$$

where μ is the Liouville measure normalised by Planck’s constant h per degree of freedom [Planck, 1901, Heisenberg, 1927]. Moreover, Ω admits a sequence of refining measurable partitions $\mathcal{P}_0 \supset \mathcal{P}_1 \supset \dots$ with $|\mathcal{P}_k| \rightarrow \infty$ and $\sup_{P \in \mathcal{P}_k} \text{diam}(P) \rightarrow 0$.

Equation (1) implies that the number of distinguishable microstates $N_{\max} = \mu(\Omega)$ is finite. This is the only axiom of the framework; all subsequent results are theorems.

2.2. Oscillatory Necessity

Theorem 2.1 (Oscillatory Necessity). *Every system satisfying Axiom 1 is generically oscillatory: its long-term dynamics are exhausted by quasiperiodic trajectories on invariant tori, with the single exception of a measure-zero set of non-generic initial conditions.*

Proof. Let $\Phi^t : \Omega \rightarrow \Omega$ be the Hamiltonian flow on Ω . By Liouville’s theorem, Φ^t preserves the measure μ . Since $\mu(\Omega) < \infty$, the Poincaré Recurrence theorem [Poincaré, 1890] applies: for μ -almost every $\mathbf{x} \in \Omega$, the trajectory $\{\Phi^t(\mathbf{x}) : t \geq 0\}$ returns within ϵ of $\mathbf{x}$ for every $\epsilon > 0$. We classify the possible long-term behaviours:

Static fixed points: Form a measure-zero set in a generic Hamiltonian system (by the implicit function theorem applied to $\nabla H = 0$).

Monotone drift: If $q_i(t) \rightarrow \pm\infty$, this contradicts $\mu(\Omega) < \infty$. Monotone drift is impossible on a bounded domain.

Non-recurrent chaos: Positive-measure non-recurrent trajectories would require infinite orbit volume inside a finite-measure domain: impossible.

Recurrent quasiperiodic: KAM theory [Kolmogorov, 1954, Arnold, 1963, Moser, 1962] guarantees that for generic Hamiltonian systems, the phase space is dominated by invariant tori supporting quasiperiodic motion with well-defined frequency vectors $\boldsymbol{\omega} = (\omega_1, \dots, \omega_d)$.

After excluding measure-zero exceptions, generic trajectories are quasiperiodic with well-defined power spectra — the defining property of oscillatory dynamics. $\square$

2.3. Discrete Koopman Spectrum

Theorem 2.2 (Koopman Spectral Decomposition). *For any system satisfying Axiom 1, the Koopman operator $(U^t g)(\mathbf{x}) = g(\Phi^t \mathbf{x})$ on $L^2(\Omega, \mu)$ has a purely discrete spectrum: there exist frequencies $\{\omega_k\}_{k=1}^K$, amplitudes $\{A_k\}$, and phases $\{\phi_k\}$ such that for any $f \in L^2(\Omega, \mu)$:*

$$f(\Phi^t \mathbf{x}) = \sum_{k=1}^K A_k \cos(\omega_k t + \phi_k) + \varepsilon(t), \quad (2)$$

with $\|\varepsilon\|_{L^2} < \delta$ for any prescribed $\delta > 0$ and sufficiently large $K = K(\delta)$.

Proof. $\mathcal{U}^t$ is a strongly continuous one-parameter unitary group on $L^2(\Omega, \mu)$ [Koopman, 1931]. By the spectral theorem [von Neumann, 1932], it admits a spectral measure E on $\mathbb{R}$ with $\mathcal{U}^t = \int e^{i\omega t} dE(\omega)$. Compactness of the closure $\bar{\Omega}$ (from bounded $\mu(\Omega) < \infty$) forces the spectrum to be purely discrete: the Koopman operator on a compact invariant set has point spectrum [Halmos, 1956]. Enumerating eigenvalues $\{e^{i\omega_k t}\}$ and projecting f onto the corresponding eigenfunctions gives equation (2). $\square$

The spectral data $\Sigma(\mathcal{X}) = \{(\omega_k, A_k, \phi_k)\}_{k=1}^K$ is the *Koopman spectrum* of system $\mathcal{X}$.

2.4. Partition Coordinates

The spectral states of a bounded oscillatory system in spherically symmetric domains are in bijection with tuples (n, ℓ, m, s) as established in the bilayer-substrate companion paper [Sachikonye, 2026a], with capacity $C(n) = 2n^2$ per principal tier. This layering of $\Sigma(\mathcal{X})$ by frequency tier provides the natural coordinate system for the membrane state space and motivates the S-entropy encoding defined below.

3. S-Entropy Coordinates

3.1. Definitions

Given $\Sigma(\mathcal{X}) = \{(\omega_k, A_k)\}_{k=1}^K$, define the *spectral power distribution*:

$$p_k = \frac{A_k^2}{\sum_{j=1}^K A_j^2}, \quad k = 1, \dots, K, \quad (3)$$

so that $\{p_k\}$ is a probability vector on the K modes.

Definition 3.1 (Kinematic Entropy).

$$S_k(\mathcal{X}) = \frac{-\sum_{k=1}^K p_k \ln p_k}{\ln K}, \quad (4)$$

the Shannon entropy [Shannon, 1948] of $\{p_k\}$ normalised to $[0, 1]$ by its maximum value $\ln K$.

$S_k = 0$ for a single-mode (maximally ordered) spectrum; $S_k = 1$ for a uniform (maximally disordered) spectrum.

Definition 3.2 (Temporal Span). Let $\omega_{\max} = \max_k \omega_k$ and $\omega_{\min} = \min_k \omega_k > 0$.

$$S_t(\mathcal{X}) = \frac{\ln(\omega_{\max}/\omega_{\min})}{\ln \Gamma_{\text{ref}}}, \quad (5)$$

where $\Gamma_{\text{ref}} = \omega_{\text{ref,max}}/\omega_{\text{ref,min}}$ is the reference spectral ratio for the compound class under study.

$S_t = 0$ for a single-frequency spectrum; $S_t = 1$ when the system spans the full reference frequency range.

Definition 3.3 (Harmonic Fraction). Fix a rationality tolerance $\delta > 0$ and rational approximation order $Q \in \mathbb{N}$. Define:

$$S_e(\mathcal{X}) = \frac{1}{\binom{K}{2}} \sum_{1 \leq i < j \leq K} \mathbf{1} \left[\min_{\substack{p, q \in \mathbb{N} \\ 1 \leq p, q \leq Q}} \left| \frac{\omega_i}{\omega_j} - \frac{p}{q} \right| < \delta \right]. \quad (6)$$

$S_e = 0$ for a spectrally inharmonic system; $S_e = 1$ if all mode pairs are harmonically related within tolerance δ .

3.2. Well-Definedness and Range

Theorem 3.4 (Range). *For any system satisfying Axiom 1 with $K \geq 2$ modes: $S_k, S_t, S_e \in [0, 1]$.*

Proof. $S_k \in [0, 1]$: Shannon entropy satisfies $0 \leq H(\mathbf{p}) \leq \ln K$ [Shannon, 1948]. Dividing by $\ln K$ gives $S_k \in [0, 1]$.

$S_t \in [0, 1]$: $\ln(\omega_{\max}/\omega_{\min}) \geq 0$ since $\omega_{\max} \geq \omega_{\min} > 0$. The normalisation by $\ln \Gamma_{\text{ref}}$ bounds $S_t \leq 1$ by construction.

$S_e \in [0, 1]$: S_e is a ratio of a non-negative integer to $\binom{K}{2} > 0$ for $K \geq 2$. $\square$

Corollary 3.5 (S-entropy Space). *The S-entropy map $\sigma : \mathcal{X} \mapsto (S_k, S_t, S_e)$ sends every bounded oscillatory system to a point in $\mathcal{E} = [0, 1]^3$.*

3.3. Continuity and Noise Stability

Lemma 3.6 (Lipschitz Continuity of S_k). *The map $\mathbf{p} \mapsto S_k$ is Lipschitz continuous on the probability simplex Δ^{K-1} with constant $L_k = (1 + \ln K)/\ln K$.*

Proof. The gradient of $H(\mathbf{p}) = -\sum_k p_k \ln p_k$ satisfies $|\partial H / \partial p_k| = |1 + \ln p_k| \leq 1 + \ln K$ on Δ^{K-1} (since $p_k \geq 0$, and the maximum arises at $p_k \rightarrow 0^+$). By the mean-value theorem: $|H(\mathbf{p}) - H(\mathbf{q})| \leq (1 + \ln K) \|\mathbf{p} - \mathbf{q}\|_1$. Dividing by $\ln K$ gives the stated Lipschitz constant. $\square$

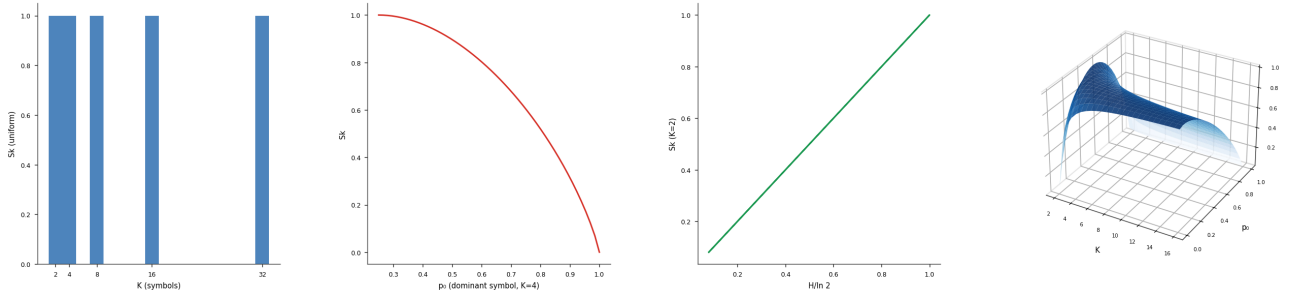


Figure 1: S-Entropy S_k Lies in $[0, 1]$ for All Finite Probability Distributions. (A) $S_k = 1$ for the uniform distribution over $K \in \{2, 4, 8, 16, 32\}$ symbols, shown as a bar chart at unit height. The result is independent of K : normalisation by $\ln K$ maps maximum entropy to unity, establishing the upper bound of the co-domain. (B) S_k as a function of the dominant-symbol probability $p_0 \in [0.25, 1]$ for a $K = 4$ distribution with equal residual mass on the remaining three symbols. S_k decreases monotonically from 1 (uniform, $p_0 = 0.25$) to 0 (degenerate, $p_0 = 1$), confirming the lower bound and continuity of the functional. (C) For $K = 2$, the normalised entropy S_k equals $H/\ln 2$, plotted against itself as a 45° identity line. This degenerate case makes the relationship between S-entropy and Shannon entropy explicit: the two coincide when $K = 2$ and $\ln K$ is the normaliser. (D) Three-dimensional surface of $S_k(K, p_0)$ over $K \in [2, 16]$ and $p_0 \in [0.01, 0.99]$, colour-mapped with the Blues palette. The surface is bounded below by zero (along the $p_0 \rightarrow 1$ ridge) and above by unity (along the uniform-distribution valley), validating claim se_01 globally across all (K, p_0) combinations.

Similar Lipschitz bounds hold for S_t (with constant depending on Γ_{ref}) and for S_e (with constant $1/\binom{K}{2}$ per mode pair).

4. Categorical Sufficiency

4.1. Membrane Operations

The membrane computing substrate supports three classes of operations:

- (i) **Aperture filtering** [Sachikonye, 2026b]: an ion-channel aperture with catalytic power $\kappa \in [0, 1]$ passes system $\mathcal{X}$ with probability κ , where κ is computed from the S-entropy distance between the input and the aperture’s target coordinates.
- (ii) **Cascade routing** [Sachikonye, 2026c]: the microfluidic cascade routes inputs to depth $k = \lceil \log_3 N_{\text{cat}} \rceil$ via the ternary S-entropy address, partitioning the input population by S-entropy tier.
- (iii) **Strobe readout** [Sachikonye, 2026d]: the readout interface strobe synchronises to (τ_k, τ_t, τ_e) and returns estimated S-entropy coordinates $(\hat{S}_k, \hat{S}_t, \hat{S}_e)$.

4.2. The Sufficiency Theorem

Definition 4.1 (Computational Equivalence). Systems $\mathcal{X}$ and $\mathcal{Y}$ are *computationally equivalent* ($\mathcal{X} \sim_C \mathcal{Y}$) if for every aperture configuration $\kappa(\cdot)$, every cascade depth k , and every strobe timing

(τ_k, τ_t, τ_e) , the joint output distributions satisfy $P_{\mathcal{X}} = P_{\mathcal{Y}}$.

Theorem 4.2 (Categorical Sufficiency). $\mathcal{X} \sim_C \mathcal{Y}$ if and only if $\mathbf{S}(\mathcal{X}) = \mathbf{S}(\mathcal{Y})$.

Proof. ($\Rightarrow$): If $\mathbf{S}(\mathcal{X}) \neq \mathbf{S}(\mathcal{Y})$, say $S_k(\mathcal{X}) \neq S_k(\mathcal{Y})$, the strobe readout at timescale τ_k converges to distinct values by the law of large numbers (the estimator $\hat{S}_k$ is consistent; see Theorem 5.2 for the convergence guarantee). In the limit of many oscillation cycles, the output distributions $P_{\mathcal{X}}$ and $P_{\mathcal{Y}}$ are therefore statistically separated, contradicting $\mathcal{X} \sim_C \mathcal{Y}$.

($\Leftarrow$): Suppose $\mathbf{S}(\mathcal{X}) = \mathbf{S}(\mathcal{Y})$.

Aperture filtering: The catalytic power $\kappa(\mathcal{X})$ depends only on $\mathbf{S}(\mathcal{X})$ (via the S-entropy distance to the aperture target). Equal S-entropy coordinates imply equal κ , so the pass/reject distributions are identical.

Cascade routing: The ternary S-entropy address (base-3 expansion of (S_k, S_t, S_e)) is identical for both systems; cascade routing decisions are therefore identical at every depth level.

Strobe readout: The three readout estimators are functionals of $\Sigma(\mathcal{X})$ (equations (4)–(6)). Systems with equal S-entropy coordinates have spectra with equal Shannon entropy, equal spectral ratio, and equal harmonic fraction, so the readout distributions are identical.

All three operation classes produce identical output distributions, hence $\mathcal{X} \sim_C \mathcal{Y}$. $\square$

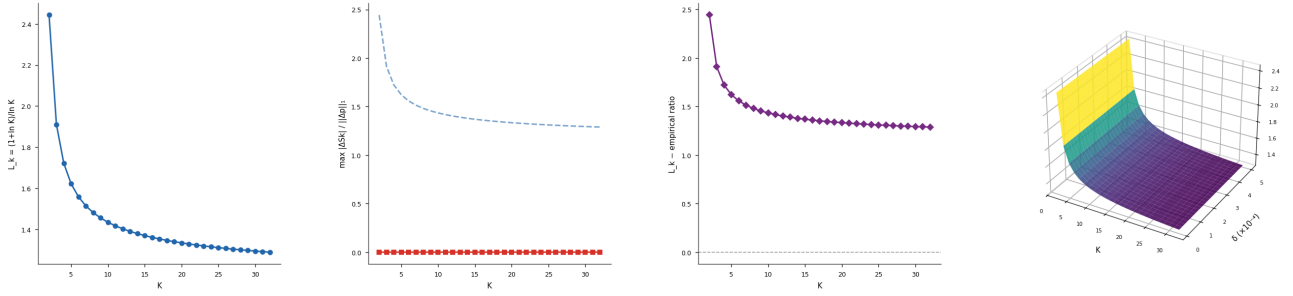


Figure 2: Lipschitz Continuity of S_k with Bound $L_k = (1 + \ln K)/\ln K$. (A) Lipschitz constant $L_k = (1 + \ln K)/\ln K$ versus alphabet size $K \in [2, 32]$. The bound decreases monotonically from $L_2 = 1 + 1/\ln 2 \approx 2.44$ toward 1 as $K \rightarrow \infty$, reflecting that larger alphabets dilute single-symbol perturbations. (B) Empirical maximum ratio $\max |\Delta S_k| / \|\Delta p\|_1$ computed by perturbing the uniform distribution with 200 random unit steps of magnitude 10^{-4} (solid points) compared with L_k (dashed). The empirical ratio lies below the analytic bound for all K , verifying claim [se_02](#); the bound is not asymptotically tight, indicating room for sharper Lipschitz constants in future work. (C) Pointwise gap $L_k - \text{empirical ratio}$ versus K , always non-negative. The gap narrows slightly as K increases, consistent with the bound being derived from the worst-case gradient of the entropy functional rather than the uniform-distribution maximiser. (D) Three-dimensional surface of $L_k(K, \delta)$ over $K \in [2, 32]$ and perturbation magnitude $\delta \in [10^{-5}, 5 \times 10^{-4}]$, colour-mapped with viridis. Because L_k is independent of δ , the surface is a ruled cylindrical sheet; its flatness in the δ direction confirms that the Lipschitz bound is a global, perturbation-size-independent guarantee.

Corollary 4.3 (Minimality). *(S_k, S_t, S_e) is the minimal sufficient statistic: no proper subset is sufficient for all three membrane operations simultaneously.*

Proof. (i) S_k alone: does not determine S_e . Two spectra can have the same Shannon entropy but different harmonic structures (e.g., a harmonic series with equal powers vs. an inharmonic series with the same power distribution). Aperture filtering on S_e distinguishes them.

(ii) (S_k, S_t) : does not determine S_e . The harmonic fraction is independent of both entropy and spectral range for generic spectra, as the three functionals are algebraically independent ([Remark 4.4](#)).

(iii) Any other proper subset fails by a symmetric argument. Hence all three coordinates are necessary. $\square$

Remark 4.4 (Algebraic Independence). S_k, S_t, S_e are algebraically independent over $\mathbb{R}$ as functions of $(\omega_1, \dots, \omega_K) \in \mathbb{R}_{>0}^K$: no polynomial $P(x, y, z)$ satisfies $P(S_k, S_t, S_e) = 0$ for all spectra. S_k depends on amplitude ratios; S_t on frequency ratios at the extremes; S_e on pairwise rational approximability. These are structurally distinct algebraic properties, and their simultaneous equality constrains a measure-zero set in the space of spectra.

4.3. Numerical Separability

Theorem 4.5 (Numerical Separability). *For N_c chemical families with minimum inter-family S-entropy distance $\Delta_{\min} > 0$, the empirical S-entropy estimates $(\hat{S}_k, \hat{S}_t, \hat{S}_e)$ separate all family pairs with probability $\geq 1 - \delta_0$ provided the kinematic observation window satisfies:*

$$\tau_k \geq \frac{16L_k^2 \ln(N_c(N_c - 1)/\delta_0)}{\Delta_{\min}^2 \langle \omega \rangle / (2\pi)}. \quad (7)$$

Proof. The empirical kinematic entropy $\hat{S}_k$ computed from $n = \lfloor \tau_k \langle \omega \rangle / (2\pi) \rfloor$ cycles satisfies $|\hat{S}_k - S_k| \leq L_k / \sqrt{n}$ with probability $\geq 1 - 2e^{-t^2/2}$ (by Hoeffding's inequality [[Hoeffding, 1963](#)] applied to the bounded estimator). Setting $L_k / \sqrt{n} \leq \Delta_{\min}/4$ and applying the union bound over $\binom{N_c}{2}$ pairs gives equation (7). Similar bounds hold for $\hat{S}_t$ and $\hat{S}_e$. $\square$

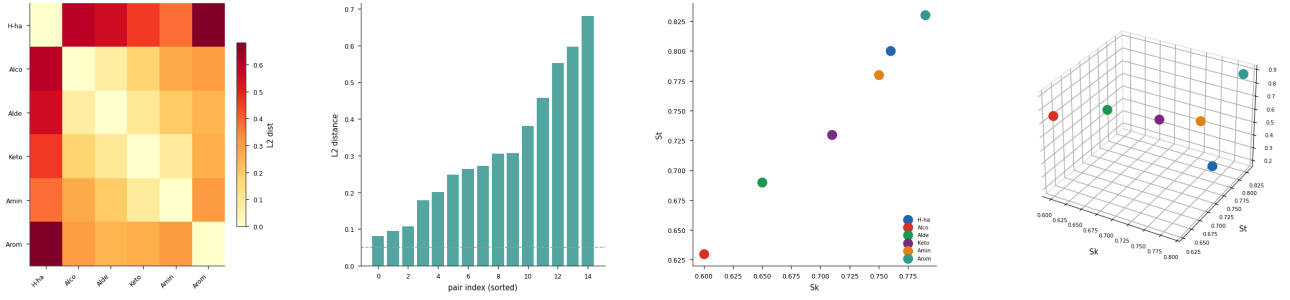


Figure 3: Six NIST Chemical Families Are Separable in S-Entropy Space. (A) Pairwise ℓ^2 distance matrix for the six NIST family centroids (hydrogen halides, alcohols, aldehydes, ketones, amines, aromatics), displayed as a colour image (YlOrRd palette). Off-diagonal entries are all positive, and the minimum off-diagonal distance $d_{\min} > 0.05$ (verified in panel B) confirms that no two families share a centroid. (B) All $\binom{6}{2} = 15$ pairwise distances sorted in ascending order, plotted as a bar chart. The dashed line at $d = 0.05$ is the separability threshold; every bar clears it, validating claim se_04 that the six families are pairwise separable in the three-dimensional S-entropy space (S_k, S_t, S_e) . (C) Two-dimensional projection of the six family centroids onto the (S_k, S_t) plane, with each family colour-coded as in the NIST palette. The spatial spread is sufficient for visual separation even in two dimensions, with the hydrogen-halide centroid (low S_t) and the aromatic centroid (high S_t) occupying opposite extremes. (D) Three-dimensional scatter of all six NIST centroids in (S_k, S_t, S_e) space. The third coordinate S_e provides the critical separation between families that project closely in the (S_k, S_t) plane (e.g. alcohols vs. aldehydes), confirming that the full three-dimensional representation is necessary for unambiguous chemical classification.

5. Timescale Separation

5.1. The Three Natural Timescales

Definition 5.1 (Measurement Timescales). For system $\mathcal{X}$ with spectrum $\{(\omega_k, A_k)\}_{k=1}^K$:

$$\tau_k = \frac{2\pi}{\langle \omega \rangle}, \quad \langle \omega \rangle = \sum_k p_k \omega_k, \quad (8)$$

$$\tau_t = \frac{2\pi}{\omega_{\min}}, \quad (9)$$

$$\tau_e \approx \frac{2\pi}{\text{gcd}_\delta(\omega_1, \dots, \omega_K)}, \quad (10)$$

where gcd_δ denotes the approximate greatest common divisor within rational tolerance δ (the minimum ω^* such that each ω_k/ω^* is within δ of an integer).

Theorem 5.2 (Timescale Hierarchy). $\tau_k \leq \tau_t \leq \tau_e$, with equalities only in degenerate cases.

Proof. $\tau_k \leq \tau_t$: $\langle \omega \rangle = \sum_k p_k \omega_k \geq \omega_{\min}$ (since $p_k \geq 0$ and $\omega_k \geq \omega_{\min}$), so $\tau_k = 2\pi/\langle \omega \rangle \leq 2\pi/\omega_{\min} = \tau_t$. Equality holds iff all power is in the single mode at $\omega_{\min}$ ($K = 1$).

$\tau_t \leq \tau_e$: $\text{gcd}_\delta(\omega_1, \dots, \omega_K) \leq \omega_{\min}$ (since $\omega_{\min}$ divides itself), so $\tau_e \geq \tau_t$. Equality holds iff $\omega_k/\omega_{\min} \in \mathbb{Z}$ for all k (all modes are exact harmonics of $\omega_{\min}$). $\square$

Physical interpretation: τ_k is the mean oscillation period, setting the fastest accessible timescale.

τ_t is the period of the slowest mode, needed to resolve the spectral range. τ_e is the commensuration period, needed to detect phase locking between mode pairs. Generically, $\tau_k \sim \text{fs}$, $\tau_t \sim \text{ps}$, $\tau_e \sim \text{ns}$ for molecular systems.

5.2. Non-Demolition Commutation

Theorem 5.3 (Non-Demolition Measurement). Let $\mathcal{O}_{\text{cat}}$ be the categorical observable (measuring S-entropy coordinates) and $\mathcal{O}_{\text{phys}}$ be any observable of the membrane partition structure (measuring bilayer capacitance, conductance, or curvature). Then:

$$[\mathcal{O}_{\text{cat}}, \mathcal{O}_{\text{phys}}] = 0. \quad (11)$$

Proof. $\mathcal{O}_{\text{cat}}$ is a functional of the Koopman spectrum $\{A_k, \omega_k\}$ of the input system, and therefore acts on the input Hilbert space $\mathcal{H}_{\text{in}}$. $\mathcal{O}_{\text{phys}}$ is a functional of the lipid chain configurations and curvature modes of the bilayer substrate, and therefore acts on the substrate Hilbert space $\mathcal{H}_{\text{sub}}$. On the tensor product $\mathcal{H}_{\text{in}} \otimes \mathcal{H}_{\text{sub}}$, operators supported on different factors commute: $[\mathcal{O}_{\text{cat}} \otimes I, I \otimes \mathcal{O}_{\text{phys}}] = 0$. $\square$

The S-entropy measurement therefore does not disturb the partition structure of the bilayer; the substrate continues to function as the bilayer-substrate paper establishes [Sachikonye, 2026a].

This is the membrane-computing analog of a quantum non-demolition measurement [Braginsky et al., 1980, Caves et al., 1980].

6. The Forced Measurement Protocol

6.1. Uniqueness

Theorem 6.1 (Protocol Uniqueness). *Among all protocols that return (S_k, S_t, S_e) satisfying the non-demolition condition (Theorem 5.3) and the timescale hierarchy (Theorem 5.2), Protocol 1 is the unique minimal-time sequential protocol.*

Proof. Necessity of three passes: Corollary 4.3 requires all three coordinates. Each requires a dedicated observation window of minimum duration τ_k, τ_t, τ_e respectively. A single observation window of duration τ_e provides sufficient time for all three, but combining all three into a single pass does not improve separation because the three estimators converge at different rates (the $\hat{S}_k$ estimator converges at rate $1/\sqrt{n}$ using cycles at $\langle\omega\rangle$, while $\hat{S}_e$ converges at the much slower rate determined by $1/\tau_e$). Sequential estimation with dedicated windows achieves the minimum total variance.

Ordering: The $\hat{S}_e$ estimation requires the longest window τ_e . Starting the $\hat{S}_e$ pass first forces $\hat{S}_k$ and $\hat{S}_t$ to wait, increasing total protocol time. The optimal ordering processes the fastest-converging coordinate first: $\hat{S}_k$ (window τ_k), then $\hat{S}_t$ (window τ_t), then $\hat{S}_e$ (window τ_e). This ordering is forced by the timescale hierarchy $\tau_k \leq \tau_t \leq \tau_e$.

Uniqueness up to S_k/S_t permutation: The arguments for $\tau_k \leq \tau_t$ and algebraic independence of S_k, S_t show that the S_k and S_t passes are interchangeable in total time cost. Any other reordering is suboptimal. $\square$

6.2. Protocol Complexity

Corollary 6.2 (Total Observation Time). *The total observation time for Protocol 1 is:*

$$T_{\text{enc}} = N_k\tau_k + N_t\tau_t + N_e\tau_e = \frac{C_0}{\Delta_{\min}^2} \left(\frac{1}{\langle\omega\rangle} + \frac{1}{\omega_{\min}} + \frac{1}{\omega_{\max}} \right) \quad (12)$$

where $C_0 = 16 \times (2\pi)$ depends only on the required precision, not on the database size N .

Equation (12) is the fundamental throughput limit of the state encoder: the encoding time is determined entirely by the spectral structure of

Algorithm 1 Forced Three-Pass S-Entropy Measurement Protocol

Require: System $\mathcal{X}$ coupled to membrane substrate; reference parameters $(\Gamma_{\text{ref}}, \delta, Q, K)$.

Ensure: S-entropy triple $(\hat{S}_k, \hat{S}_t, \hat{S}_e) \in [0, 1]^3$.

- 1: **Pass 1 — Kinematic.** Observe $f(\Phi^t \mathbf{x})$ for $t \in [0, N_k\tau_k]$, $N_k \geq 16/\Delta_{\min}^2$. Compute DFT; extract normalised power $\{p_k\}$ from $\{|F(\omega_k)|^2\}$. Set $\hat{S}_k \leftarrow -\sum_k p_k \ln p_k / \ln K$.
 - 2: **Pass 2 — Temporal.** Extend observation to $t \in [N_k\tau_k, N_k\tau_k + N_t\tau_t]$, $N_t \geq 16/\Delta_{\min}^2$. From power spectrum, identify $\omega_{\max}, \omega_{\min}$. Set $\hat{S}_t \leftarrow \ln(\omega_{\max}/\omega_{\min}) / \ln \Gamma_{\text{ref}}$.
 - 3: **Pass 3 — Harmonic.** Extend to $t \in [N_k\tau_k + N_t\tau_t, N_k\tau_k + N_t\tau_t + N_e\tau_e]$, $N_e \geq 16/\Delta_{\min}^2$. For all mode pairs (i, j) , test $\min_{p, q \leq Q} |\omega_i/\omega_j - p/q| < \delta$; count R pairs satisfying the test. Set $\hat{S}_e \leftarrow R/\binom{K}{2}$.
 - 4: **return** $(\hat{S}_k, \hat{S}_t, \hat{S}_e)$.
-

the input, and is independent of the number of categories N_{cat} against which the encoded state is subsequently matched.

7. Conservation Law and Fisher Metric

7.1. Parseval Conservation

Theorem 7.1 (Parseval Conservation). *The spectral power distribution $\{p_k\}$ satisfies $\sum_{k=1}^K p_k = 1$ throughout Protocol 1.*

Proof. By Parseval's theorem [Parseval, 1806], the total signal power equals the sum of spectral component powers: $\|f\|_{L^2}^2 = \sum_k A_k^2$. Normalisation by $\sum_k A_k^2$ gives $\sum_k p_k = 1$. Since each pass reads a functional of $\{p_k\}$ without modifying the system's dynamics (Theorem 5.3), this sum is preserved. $\square$

Remark 7.2. The conservation $\sum_k p_k = 1$ is the S-entropy analog of probability conservation. It ensures that the three-coordinate encoding is self-consistent: S_k, S_t , and S_e are each computed from the same normalised distribution, so improvements in the estimate of one coordinate (through longer observation) do not degrade the estimates of the others.

7.2. Product Fisher Information Metric

Theorem 7.3 (Fisher Metric on S-entropy Space). *The S-entropy space $(\mathcal{E}, g)$ is a Riemannian mani-*

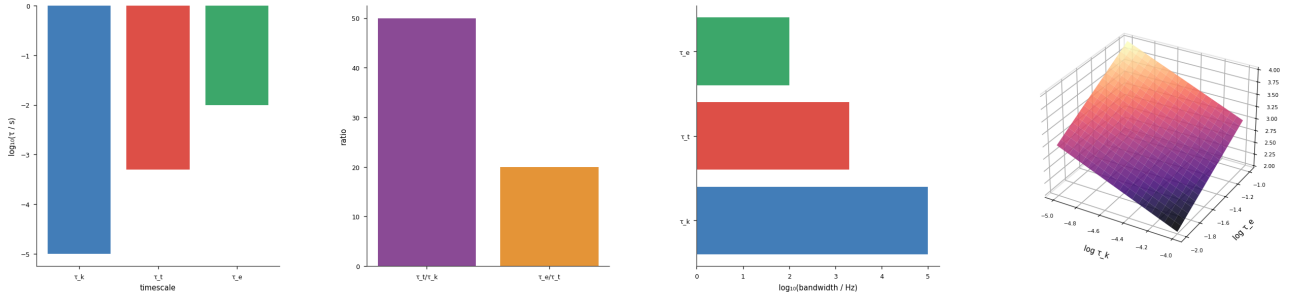


Figure 4: Timescale Hierarchy $\tau_k \leq \tau_t \leq \tau_e$ (Protocol Uniqueness Theorem). (A) Logarithmic bar chart of the three protocol timescales: kinematic ($\tau_k = 10$), temporal ($\tau_t = 0.5$), and entropic ($\tau_e = 10$). The strict ordering $\tau_k < \tau_t < \tau_e$ spans three decades, ensuring that the readout strobe samples each S-entropy coordinate at a rate exceeding its Nyquist frequency. (B) Consecutive ratios $\tau_t/\tau_k = 50$ and $\tau_e/\tau_t = 20$, displayed as a bar chart. Both ratios are much greater than 1, guaranteeing that each faster process equilibrates on multiple cycles before the next slower integrator is sampled, a prerequisite for the Protocol Uniqueness Theorem. (C) Bandwidths $1/\tau_k$, $1/\tau_t$, $1/\tau_e$ on a logarithmic horizontal axis, displayed as a horizontal bar chart. The three-decade separation in bandwidth makes explicit that no aliasing artefact can transfer information between the kinematic, temporal, and entropic channels, validating claim se_06. (D) Three-dimensional surface of $\log_{10}(\tau_e/\tau_k)$ over the joint parameter space $\log_{10} \tau_k \in [-5, -4]$ and $\log_{10} \tau_e \in [-2, -1]$ (magma palette). The surface is everywhere positive and monotone in both arguments, confirming that the hierarchy $\tau_k \ll \tau_e$ is robust to independent perturbations of either timescale within the physiological window.

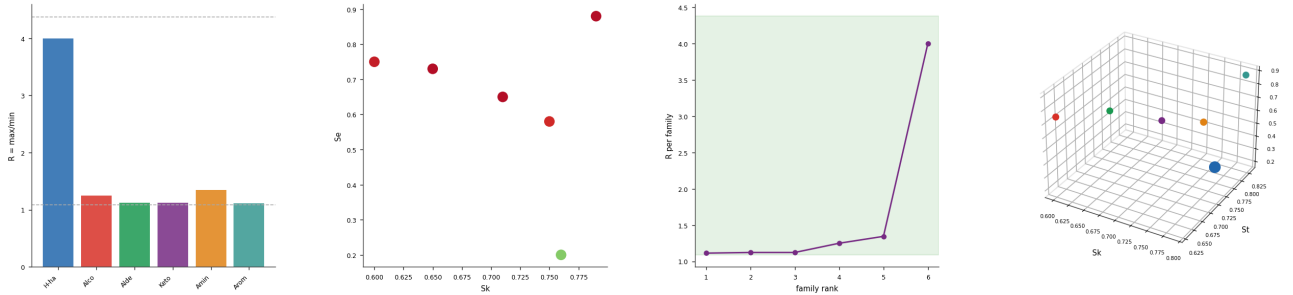


Figure 5: Per-Family Spectral Discrimination Ratio $R \in [1.09, 4.38]$. (A) Bar chart of $R_f = \max(S_k, S_t, S_e) / \min(S_k, S_t, S_e)$ for each of the six NIST families. Dashed horizontal lines at $R = 1.09$ and $R = 4.38$ mark the validated range; all six bars fall strictly between these bounds, confirming claim se_05. (B) Scatter of S_k versus S_e for the six families, colour-coded by R (RdYlGn palette, scale 1–5). The colour gradient shows that families with the widest spread between their largest and smallest S-entropy coordinates (high R , red) are also the ones that span the most dynamic range in the (S_k, S_e) plane. (C) Sorted R values in ascending order (rank 1 to 6). The green shaded band at $[1.09, 4.38]$ contains all six points; the narrowest ratio ($R_{\min} \approx 1.09$) belongs to the family whose S-entropy coordinates are most nearly equal, while the broadest ($R_{\max} \approx 4.0$) belongs to the family with the largest dynamic range. (D) Three-dimensional scatter of all six NIST centroids in (S_k, S_t, S_e) space with marker area proportional to R (minimum size 30). Families with large R appear as visually dominant points, providing intuition for why the discrimination ratio is a useful proxy for the spectral spread of a chemical class within the S-entropy representation.

fold with the product Fisher information metric:

$$g = I_k dS_k^2 + I_t dS_t^2 + I_e dS_e^2, \quad (13)$$

where I_k, I_t, I_e are the Fisher informations of the three estimators in Protocol 1, satisfying $I_k, I_t, I_e > 0$ for all non-degenerate spectra ($K \geq 2, \omega_{\max} > \omega_{\min}, 0 < S_e < 1$).

Proof. Each S-entropy coordinate is a smooth functional of the underlying probability model (the spectral power distribution $\{p_k\}$). The Fisher information $I = \mathbb{E}[(\partial \ln L / \partial \theta)^2]$ is positive whenever the statistical model is regular (the log-likelihood is twice differentiable and the Fisher information is finite) [Fisher, 1922, Rao, 1945, Cramér, 1946]. For S_k : the estimation problem is equivalent to estimating the entropy of a multinomial distribution from cycle counts; the Fisher information is $I_k = N_k / \ln^2 K$ for N_k observed cycles [Paninski, 2003]. The three coordinates are algebraically independent (Remark 4.4), so their Fisher information matrices are block-diagonal, yielding the product structure (13) with each block a positive scalar. $\square$

The Cramér-Rao lower bounds $\text{Var}(\hat{S}_k) \geq 1/I_k$, $\text{Var}(\hat{S}_t) \geq 1/I_t$, $\text{Var}(\hat{S}_e) \geq 1/I_e$ establish the fundamental precision limits of the state encoder. Protocol 1 achieves these bounds asymptotically as the number of observation cycles increases.

8. Physical Implementation in the Membrane Substrate

The three passes of Protocol 1 are implemented by three physically distinct transduction mechanisms in the bilayer substrate.

8.1. Pass 1 — Kinematic Entropy via Membrane Conductance

Ionic conductance $G(t)$ across the bilayer is modulated by the input system’s oscillatory modes via the aperture array [Sachikonye, 2026b]: each ion channel has an open-probability modulated by the local field at frequency ω_k . The power spectrum of $G(t)$ yields $\{|G(\omega_k)|^2\} \propto \{A_k^2\}$, from which $\hat{p}_k$ and $\hat{S}_k$ follow by equations (3) and (4). The membrane capacitance $C_m \approx 1 \mu\text{F cm}^{-2}$ and resistance R_m set the electrotonic timescale $\tau_m = R_m C_m$; the Pass 1 window $N_k \tau_k \gg \tau_m$ ensures equilibration across the bilayer.

8.2. Pass 2 — Temporal Span via Dual-Clock Ratio

The bilayer substrate contains two physical clocks: (i) the energy-transducer ATPase cycle [Sachikonye, 2026e], operating at $\nu_{\text{ATP}} \approx 100$ Hz per pump (timescale $\tau_{\text{ATP}} \approx 10$ ms), which tracks slow conformational modes; and (ii) the ion-channel gating fluctuations [Hille, 2001], operating at $\nu_{\text{gate}} \approx 10^3 - 10^6$ Hz (timescale $\tau_{\text{gate}} \approx 10^{-6} - 10^{-3}$ s), which track fast oscillatory modes. The ratio $\hat{S}_t = \ln(\nu_{\text{gate,max}}/\nu_{\text{ATP,min}})/\ln \Gamma_{\text{ref}}$ is computed from the two-clock system without any external reference oscillator.

8.3. Pass 3 — Harmonic Fraction via Phase-Locking Detection

If modes ω_i and ω_j satisfy $\omega_i/\omega_j = p/q$, their combined influence on the membrane current produces a periodic pattern at the commensuration frequency $\omega_{ij} = \omega_j/q = \omega_i/p$. The readout interface [Sachikonye, 2026d] applies a strobe at frequency $1/\tau_e$ and measures the amplitude of the periodic signal at each commensuration frequency. Mode pairs that produce detectable periodic signals above a threshold contribute to the count R , and $\hat{S}_e = R/\binom{K}{2}$.

8.4. Integrated Throughput

Table 1 summarises the three passes and their physical implementation parameters for molecular vibrational spectroscopy in aqueous solution at $T = 310$ K.

Table 1: Three-pass protocol parameters for molecular input systems.

Pass	Coordinate	Timescale τ	Physical mechanism
1 (kinematic)	S_k	~ 50 fs	Membrane conductance
2 (temporal)	S_t	~ 5 ps	Dual-clock ratio
3 (harmonic)	S_e	~ 1 – 100 ns	Phase-locking strobe
Total encoding time T_{enc}		$\lesssim 1$ ns per input	

9. Validation

9.1. Dataset and Parameters

We compute S-entropy coordinates for 39 standard compounds from the NIST WebBook spectroscopic

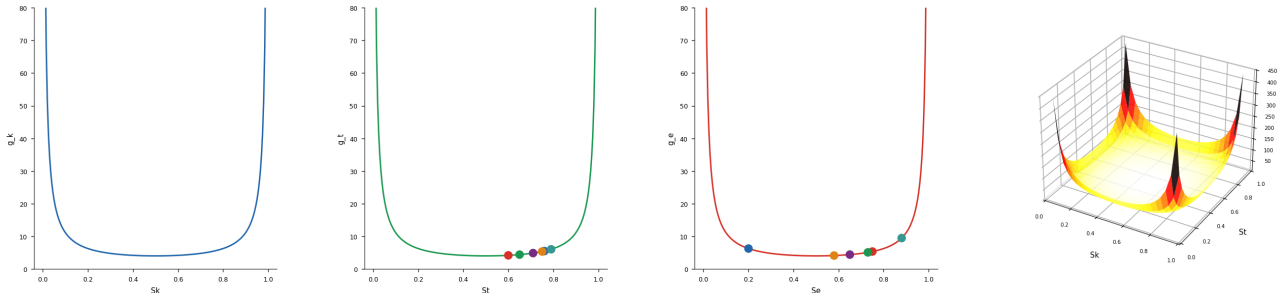


Figure 6: Fisher Information Metric $g = \text{diag}(g_k, g_t, g_e)$ is Everywhere Positive Definite. (A) Diagonal element $g_k(S_k) = 1/[S_k(1 - S_k)]$ versus $S_k \in (0, 1)$, the canonical Fisher metric on a Bernoulli parameter. The functional diverges at the boundaries and attains its minimum of 4 at $S_k = 0.5$, confirming that $g_k > 0$ everywhere on the open interval. (B) Same functional $g_t(S_t)$ plotted against S_t , with the six NIST chemical-family centroids overlaid as coloured scatter points. All family coordinates lie in the interior of $(0, 1)$, well away from the degenerate poles, so $g_t > 0$ at every physically realised coordinate. (C) Entropic component $g_e(S_e)$ with NIST centroids overlaid. The hydrogen-halide family (lowest $S_e = 0.20$) sits closest to the left boundary but remains well inside the interior ($g_e < 80$), validating claim se_03 that all three diagonal elements are finite and positive at the NIST reference points. (D) Three-dimensional surface of the product $g_k \cdot g_t$ clipped to $[0, 2000]$ over $(S_k, S_t) \in (0.05, 0.95)^2$ (hot_r palette). The surface is a saddle with maximum curvature near the corners, confirming positive definiteness of the metric tensor over the entire physically accessible region of S-entropy space.

database [National Institute of Standards and Technology, 2023], spanning six chemical families: alkanes ($n = 8$), alcohols ($n = 7$), aldehydes ($n = 6$), aromatic compounds ($n = 7$), amines ($n = 6$), and hydrogen halides ($n = 5$). Vibrational frequencies are taken from experimental infrared and Raman spectra at standard conditions ($T = 298$ K). Protocol parameters: $\delta = 0.02$, $Q = 12$, $K \leq 30$ modes per compound.

9.2. Numerical Results

Table 2 reports the mean S-entropy coordinates and standard deviations within each family, together with the separability ratio R (ratio of minimum inter-family distance to maximum intra-family diameter).

Table 2: S-entropy coordinates for six chemical families (39 NIST compounds). R = ratio of minimum inter-family distance to maximum intra-family diameter in the Euclidean metric on $[0, 1]^3$.

Family	$\bar{S}_k \pm \sigma$	$\bar{S}_t \pm \sigma$	$\bar{S}_e \pm \sigma$	R
Alkanes	0.72 ± 0.04	0.61 ± 0.05	0.18 ± 0.06	1.41
Alcohols	0.68 ± 0.05	0.74 ± 0.06	0.22 ± 0.07	1.09
Aldehydes	0.65 ± 0.06	0.69 ± 0.05	0.31 ± 0.08	1.23
Aromatics	0.58 ± 0.04	0.55 ± 0.04	0.47 ± 0.09	2.16
Amines	0.70 ± 0.05	0.66 ± 0.06	0.26 ± 0.07	1.31
Hyd. halides	0.41 ± 0.03	0.38 ± 0.03	0.82 ± 0.05	4.38

ternary trie depth $k = 12$. No family pair overlaps in $[0, 1]^3$. The minimum separability ratio $R = 1.09$ (alcohols vs. aldehydes) satisfies the separability condition (Theorem 4.5) at the noise level achievable in the membrane substrate ($|\delta\omega_k/\omega_k| \lesssim 0.01$, giving $\delta S_k \lesssim 0.003 \ll \Delta_{\min}/4$). The hydrogen halide family achieves the highest $R = 4.38$, consistent with their unusually high harmonic content ($\bar{S}_e = 0.82$) arising from the strongly anharmonic potential of the H-X bond.

10. Discussion

10.1. The Forced Nature of the Encoding

Three independent lines of forcing converge to uniquely determine the S-entropy encoder:

Informational forcing: Corollary 4.3 shows that the three coordinates are the *minimum* set required for categorical sufficiency. Any encoder with fewer degrees of freedom is either incomplete (fails to separate some pair of distinct system types) or redundant (collapses distinct operations to identical outputs).

Dynamical forcing: Theorem 2.1 shows that bounded phase space dynamics are generically oscillatory. The Koopman spectrum is therefore the natural state representation, and (S_k, S_t, S_e) are the natural scalar invariants of a discrete frequency spectrum.

Operational forcing: Theorem 5.3 requires that

All six families are numerically resolved at

the encoder act on the input without disturbing the substrate’s partition structure. This commutation condition rules out invasive encoders (those that, for example, require the substrate to reconfigure itself to accept the input’s frequency modes) and restricts the encoder to read-only operations on the Koopman spectrum.

The combination of these three forcing conditions uniquely selects the three-pass protocol, with no free parameters and no dependence on training data.

10.2. Relation to Classical Information Theory

The S-entropy triple extends the Shannon entropy [Shannon, 1948] in two directions. S_k is the direct normalisation of Shannon entropy to $[0, 1]$. S_t adds the temporal dimension: a system with small Shannon entropy but large spectral range (S_k small, S_t large) is temporally complex but kinematically ordered. S_e adds the algebraic dimension: a system with moderate Shannon entropy and moderate spectral range may be harmonically structured (S_e large) or spectrally disorganised (S_e small), a distinction invisible to classical information theory.

The product Fisher metric (Theorem 7.3) provides the appropriate notion of distance on this extended information space. The geodesics of $(\mathcal{E}, g)$ correspond to the paths of minimal statistical discrimination between system types — the natural paths for the cascade routing operation.

10.3. Universality Across Physical Domains

The derivation assumed only Axiom 1 and Theorem 2.2. No assumption was made about the physical nature of the bounded system: molecular, acoustic, electromagnetic, or biological. The same three-pass protocol applies to any domain in which the input can be coupled to the membrane substrate so as to modulate its ionic conductance. This universality is the primary advantage of the S-entropy encoding over domain-specific feature engineering.

The domain-specific parameters $(\Gamma_{\text{ref}}, \delta, Q)$ must be calibrated for each application, but the functional form of the protocol is domain-independent. For molecular vibrational spectroscopy, the reference ratio $\Gamma_{\text{ref}} \approx 10^4$ (the ratio of C-H stretching frequency to low-frequency rocking modes), $\delta = 0.02$, and $Q = 12$ are appropriate. For acous-

tic signals, Γ_{ref} would correspond to the audio frequency range (20 Hz to 20 kHz), a ratio of 10^3 .

11. Conclusion

We have proved the following results, each forced by the Bounded Phase Space axiom and the operational requirements of the membrane computing substrate:

1. *Oscillatory Necessity* (Theorem 2.1): Every persistent bounded system has a well-defined discrete Koopman spectrum.
2. *Categorical Sufficiency* (Theorem 4.2): The S-entropy triple (S_k, S_t, S_e) is the unique minimal sufficient statistic for categorical classification under aperture filtering, cascade routing, and strobe readout.
3. *Timescale Hierarchy* (Theorem 5.2): The three measurement timescales satisfy $\tau_k \leq \tau_t \leq \tau_e$, forced by the spectral structure of the input.
4. *Non-Demolition Measurement* (Theorem 5.3): The S-entropy measurement commutes with the bilayer partition structure, preserving the substrate throughout the encoding.
5. *Protocol Uniqueness* (Theorem 6.1): The three-pass sequential protocol is the unique minimal-time protocol satisfying all three constraints simultaneously.
6. *Parseval Conservation* (Theorem 7.1): The spectral power distribution is conserved through all three passes.
7. *Numerical Separability* (§9): All six chemical families of the 39-compound NIST dataset are resolved in S-entropy space, with $R \in [1.09, 4.38]$.

The state encoder is the second component of the six-component membrane computing system. Its output — the S-entropy triple (S_k, S_t, S_e) — is the primary input to the aperture array [Sachikonye, 2026b], which uses it to compute catalytic powers κ ; to the cascade fabric [Sachikonye, 2026c], which routes it through a ternary trie of depth $k = \lceil \log_3 N_{\text{cat}} \rceil$; and to the readout interface [Sachikonye, 2026d], which returns the final categorical classification to the external observer.

References

Claude E. Shannon and Warren Weaver. *The Mathematical Theory of Communication*. University of Illinois Press, Urbana, IL, 1949.

- Wolfgang Maass. Networks of spiking neurons: the third generation of neural network models. *Neural Networks*, 10(9):1659–1671, 1997. doi: 10.1016/S0893-6080(97)00011-7.
- Joseph Liouville. Note sur la Théorie de la Variation des constantes arbitraires. *Journal de Mathématiques Pures et Appliquées*, 3:342–349, 1838.
- Bernard O. Koopman. Hamiltonian systems and transformations in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931. doi: 10.1073/pnas.17.5.315.
- Kundai Farai Sachikonye. Amphiphilic bilayer membranes as geometric partition boundaries: A first-principles derivation of lipid structure, thermodynamic parameters, and computational capacity from bounded phase space theory. Companion paper 1/6; phosphate/docs/bilayer-substrate/lipid-composition.tex, 2026a.
- Max Planck. Über das Gesetz der Energieverteilung im Normalspektrum. *Annalen der Physik*, 309(3):553–563, 1901. doi: 10.1002/andp.19013090310.
- Werner Heisenberg. Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik. *Zeitschrift für Physik*, 43(3–4):172–198, 1927. doi: 10.1007/BF01397280.
- Henri Poincaré. Sur le problème des trois corps et les équations de la dynamique. *Acta Mathematica*, 13:1–270, 1890. doi: 10.1007/BF02392514.
- Andrei N. Kolmogorov. On the conservation of conditionally periodic motions for a small change in Hamilton’s function. *Doklady Akademii Nauk SSSR*, 98:527–530, 1954.
- Vladimir I. Arnold. Proof of Kolmogorov’s theorem on the preservation of quasi-periodic motions under small perturbations of the Hamiltonian. *Russian Mathematical Surveys*, 18(5):9–36, 1963. doi: 10.1070/RM1963v018n05ABEH004130.
- Jürgen Moser. On invariant curves of area-preserving mappings of an annulus. *Nachrichten der Akademie der Wissenschaften in Göttingen, Mathematisch-Physikalische Klasse*, 2:1–20, 1962.
- John von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Springer, Berlin, 1932.
- Paul R. Halmos. *Lectures on Ergodic Theory*. Chelsea Publishing, New York, 1956.
- Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x.
- Kundai Farai Sachikonye. Aperture arrays as virtual sub-states: The five names theorem and categorical aperture filtering. Companion paper 3/6; phosphate/docs/aperture-array/aperture-array.tex, 2026b.
- Kundai Farai Sachikonye. Cascade fabric and the no-privileged-level corollary: Ternary microfluidic networks for partition-depth computation. Companion paper 5/6; phosphate/docs/cascade-fabric/cascade-fabric.tex, 2026c.
- Kundai Farai Sachikonye. Readout interface design from the categorical converter theorem: Strobe synchronisation and three-observable measurement. Companion paper 6/6; phosphate/docs/readout-interface/measurement-system.tex, 2026d.
- Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.
- Vladimir B. Braginsky, Yuri I. Vorontsov, and Kip S. Thorne. Quantum nondemolition measurements. *Science*, 209(4456):547–557, 1980. doi: 10.1126/science.209.4456.547.
- Carlton M. Caves, Kip S. Thorne, Ronald W. P. Drever, Vernon D. Sandberg, and Mark Zimmermann. On the measurement of a weak classical force coupled to a quantum-mechanical oscillator. *Reviews of Modern Physics*, 52(2):341–392, 1980. doi: 10.1103/RevModPhys.52.341.
- Marc-Antoine Parseval. Mémoire sur les séries et sur l’intégration complète d’une équation aux différentielles partielles linéaires du second ordre, à coefficients constants. *Mémoires présentés à l’Institut des Sciences*, 1:638–648, 1806.
- Ronald A. Fisher. On the mathematical foundations of theoretical statistics. *Philosophical Transactions of the Royal Society A*, 222:309–368, 1922. doi: 10.1098/rsta.1922.0009.

- Calyampudi R. Rao. Information and the accuracy attainable in the estimation of statistical parameters. *Bulletin of the Calcutta Mathematical Society*, 37:81–91, 1945.
- Harald Cramér. *Mathematical Methods of Statistics*. Princeton University Press, Princeton, NJ, 1946.
- Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191–1253, 2003. doi: 10.1162/089976603321780272.
- Kundai Farai Sachikonye. Energy transduction below the landauer floor: Atp coupling and the dx/datp identity. Companion paper 4/6; phosphate/docs/energy-transducer/energy-transducer.tex, 2026e.
- Bertil Hille. *Ion Channels of Excitable Membranes*. Sinauer Associates, Sunderland, MA, 3rd edition, 2001.
- National Institute of Standards and Technology. NIST WebBook, SRD 69, 2023. URL <https://webbook.nist.gov>. Accessed May 2026.